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Biosimilars versus the originator of follitropin alfa for ovarian stimulation in ART: a systematic review and meta-analysis

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ABSTRACT

STUDY QUESTION: Is the probability of pregnancy different between women using biosimilars versus the originator of follitropin alfa for ovarian stimulation in ART?

SUMMARY ANSWER: Meta-analysis of eight randomized clinical trials (RCTs) suggests that live birth, clinical, and ongoing pregnancy rates are significantly lower with biosimilars of follitropin alfa compared to the originator.

WHAT IS KNOWN ALREADY: All biosimilars of follitropin alfa have received regulatory approval by demonstrating non-inferiority in the number of retrieved oocytes compared to the originator. Nevertheless, the most clinically relevant outcome in ART for both clinicians and patients is live birth. A meta-analysis published in 2021 suggested that biosimilars of follitropin alfa are associated with lower live birth rates compared to the originator. Since then, more relevant RCTs have been published, and thus an updated critical synthesis of the available evidence is urgently warranted.

STUDY DESIGN, SIZE, DURATION: A systematic review and meta-analysis were performed to compare biosimilars versus the originator of follitropin alfa in women undergoing ovarian stimulation for ART. A literature search was conducted until January 2024 in MEDLINE, Embase, Cochrane CENTRAL, Scopus, Web of Science, WHO, Clinicaltrials.gov, and others to identify eligible RCTs. The primary outcome was live birth. Secondary outcomes included clinical and ongoing pregnancy, duration of gonadotrophin administration and total FSH dose, number of oocytes retrieved, and ovarian hyperstimulation syndrome (OHSS).

PARTICIPANTS/MATERIALS, SETTING, METHODS: Data were extracted independently by two reviewers. Quality was assessed using the RoB-2 Tool by Cochrane, and a sensitivity analysis was performed by excluding studies having high risk of bias. Meta-analysis was performed using the random or fixed effects model depending on the presence or not of significant (>50%) statistical heterogeneity (I^2). Results were combined using the intention-to-treat principle and are reported as risk ratio (RR) or weighted-mean-difference (WMD) with 95% CIs.

MAIN RESULTS AND THE ROLE OF CHANCE: Eight RCTs (n = 2987) (published between 2015 and 2023) were identified, assessing seven biosimilar products of follitropin alfa. The number of patients included in the eligible studies ranged from 100 to 1100. Three of the RCTs were deemed to be at high risk of bias. The duration of gonadotrophin administration was shorter in the biosimilars group (WMD: –0.19 days, 95% CI: –0.34 to –0.05; $I^2 = 0\%$, 5 studies, n = 2081), while no difference was observed in the total dose of FSH (WMD: –34.69 IUs, 95% CI: –74.54 to 5.16; $I^2 = 15.53\%$, 5 studies, $n = 2081$). No difference was observed in the number of oocytes retrieved (WMD: 0.27, 95% CI: –0.43 to 0.96; $I^2 = 10.7\%$, 6 studies, n = 1527) and OHSS rates (RR: 1.17, 95% CI: 0.90–1.52; $I^2 = 0\%$, 8 studies, n = 2986) between the two groups. A significantly lower live birth rate was observed using the biosimilars of follitropin alfa compared to the originator in women undergoing ovarian stimulation for ART (RR: 0.83, 95% CI: 0.72–0.96; $I^2 = 0\%$, 6 studies, n = 2335; moderate certainty of evidence). Similarly, clinical pregnancy (RR: 0.82, 95% CI: 0.73–0.92; $I^2 = 0\%$, 7 studies, n = 2876; low certainty of evidence) and ongoing pregnancy rates (RR: 0.81, 95% CI: 0.70–0.94; $I^2 = 0\%$, 7 studies, n = 1886; low certainty of evidence) were lower in the biosimilars group. These results were not materially altered in the sensitivity analyses performed where studies deemed at high risk of bias were excluded.

LIMITATIONS, REASONS FOR CAUTION: This meta-analysis included RCTs evaluating seven different biosimilars of follitropin alfa; however, pooled data appeared to be homogeneous. No data were available comparing biosimilars of follitropin alfa with the originator regarding cumulative live birth rate per aspiration or the probability of live birth in frozen thawed cycles. The population examined in the eligible RCTs includes mainly normal responders and no RCTs were identified focusing on poor or high responders.

WIDER IMPLICATIONS OF THE FINDINGS: Clinicians should be informed that although biosimilars of follitropin alfa produce similar number of oocytes with the originator, pregnancy rates after a fresh transfer are likely to be lower. Future research should focus on

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optimizing the production and use of biosimilars of follitropin alfa, so that they lead to pregnancy rates comparable to the originator.

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REGISTRATION NUMBER: Protocol for the systematic review registered in The International Prospective Register of Systematic Reviews (PROSPERO; CRD42024498237).

Keywords: biosimilars / r-hFSH / follitropin-alfa / IVF / ICSI / live birth rate

Introduction

Ovarian stimulation represents an integral part of ART, as it has been shown that obtaining multiple oocytes is associated with a higher number of good-quality embryos, live birth, and cumulative live birth rates (LBRs) (Law *et al.*, 2019; Venetis *et al.*, 2019; Vermey *et al.*, 2019; Law *et al.*, 2021; Zhang *et al.*, 2021). The first recombinant human FSH molecule (reference product recombinant human follicle-stimulating hormone, rhFSH, follitropin alfa) received marketing approval in Europe in 1995 (GONAL-f®, Merck KGaA, Darmstadt, Germany) (European Medicines Agency, 2010) and in the USA in 1997 (GONAL-f® RFF, EMD Serono, Inc., Rockland, MA) (Food and Drug Administration, 2013). Biosimilar preparations are defined as biological medicinal products that contain a version of the active substance of an already authorized original biological medicinal product (reference medicinal product). Similarity to the reference medicinal product in terms of quality characteristics, biological activity, safety, and efficacy based on a comprehensive comparability exercise needs to be established (European Medicines Agency, 2014). Two biosimilars of follitropin alfa have been approved in the EU, Ovaleap (Strowitzki *et al.*, 2016) (Theramex, Ireland; launched in 2013) and Bemfola® (Rettenbacher *et al.*, 2015) (Gedeon Richter PLC, Hungary; launched in 2014; also known as Afolia in the USA (Fertility Biotech AG, 2017)), based on Phase III clinical studies and by demonstrating non-inferiority to the reference product Gonal-F in the number of oocytes retrieved. Similarly, Primapur (Barakhoeva *et al.*, 2019) in Russia, Follitrope (Hu *et al.*, 2020) and QL1012 (Hu *et al.*, 2023a) in China, Cinnal-f (Rashidi *et al.*, 2021) in Iran, and Folitime (Pasqualini *et al.*, 2021) in Argentina have all received approval from the local authorities and were launched in the respective countries, by demonstrating comparability regarding the number of oocytes retrieved.

Nevertheless, the most clinically relevant outcome in ART for both clinicians and patients is live birth (Min *et al.*, 2004; Malizia *et al.*, 2009; Braam *et al.*, 2018), which is supported to be well correlated with clinical and ongoing pregnancy as possible secondary clinical outcomes in randomized clinical trials (RCTs) (Clarke *et al.*, 2010; Braakhekke *et al.*, 2014). The Phase III clinical trials of the aforementioned biosimilar preparations were powered to detect a difference in the number of oocytes retrieved as the primary endpoint, but also reported outcomes for live birth (Rettenbacher *et al.*, 2015; Strowitzki *et al.*, 2016; Fertility Biotech AG, 2017; Barakhoeva *et al.*, 2019; Pasqualini *et al.*, 2021; Hu *et al.*, 2023a), clinical (Rettenbacher *et al.*, 2015; Strowitzki *et al.*, 2016; Fertility Biotech AG, 2017; Hu *et al.*, 2020; Pasqualini *et al.*, 2021; Rashidi *et al.*, 2021; Hu *et al.*, 2023a), and ongoing pregnancy (Rettenbacher *et al.*, 2015; Strowitzki *et al.*, 2016; Barakhoeva *et al.*, 2019; Hu *et al.*, 2020; Pasqualini *et al.*, 2021; Rashidi *et al.*, 2021; Hu *et al.*, 2023a). However, these studies were not powered to detect differences in these outcomes.

Systematic reviews and meta-analyses are widely regarded as the optimal approach to synthesize outcomes from various trials, particularly when reduced power of individual studies is suspected. A meta-analysis published in 2021 (Chua *et al.*, 2021) suggested that biosimilars of follitropin alfa are associated with lower LBRs compared to the originator. Since then, more relevant RCTs have been published, and thus an updated critical synthesis of the available evidence is urgently warranted.

Materials and methods

Study protocol

The Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) was used during the conduct of this study (http://www.prisma-statement.org). A protocol for the systematic review was registered in The International Prospective Register of Systematic reviews (PROSPERO; CRD42024498237) prior to the initiation of the literature search.

Literature searches

Two independent reviewers performed a computerized literature search in MEDLINE, Embase, CENTRAL (The Cochrane Library), Scopus, Web of Science, Clinicaltrials.gov, US Food and Drug Administration (FDA), European Medicines Agency (EMA), and the World Health Organization (WHO) registries, published up to January 2024, in order to identify all the available RCTs comparing biosimilars versus the originator of follitropin alfa in women undergoing ovarian stimulation for ART. Relevant keywords/terms, synonyms, related variants, and database-specific indexing terminology were used to describe the biosimilar preparations of rhFSH and the reference product, the intervention, the study type, and the population (Supplementary Table S1). Additionally, the citation lists of relevant publications and systematic reviews were hand-searched.

Study selection

The studies retrieved by the literature search were screened independently by two reviewers (K.I.K. and C.A.V.) based on titles, abstracts, and then by full text. Any disagreement was resolved by discussion.

The inclusion and exclusion criteria were set prior to the literature search as follows: the studies had to be RCTs comparing biosimilars of follitropin alfa with the reference product (Gonal-f) in couples with infertility of any type and duration, undergoing controlled ovarian stimulation for IVF/ICSI in fresh and/or frozen–thawed cycles. Studies involving asymmetric co-interventions (i.e. trials in which the protocols for both treatment arms were not the same, except for the intervention assessed) were excluded.

Furthermore, pseudo RCTs were also not considered eligible for the present systematic review and meta-analysis.

Data collection

Demographic data were extracted from each of the eligible studies, including the year and the country of publication, as well as the size of the study population. Information regarding the study design was also reported, such as the methodological steps for the randomization and the allocation concealment, the blinding, whether power analysis was performed, whether there was financial support for the trial, and whether there was a protocol registration (Table 1). With regard to the actual trial and protocols details, information was extracted in a predefined standardized data extraction form assessing population characteristics and details of ART treatment protocol used for the randomized controlled trials included in the meta-analysis (Table 2). The data collection was performed independently by two authors, and any disagreement was resolved by a third party (A.S.). In accordance with the Cochrane Handbook for Systematic Reviews of Interventions (Higgins *et al.*, 2024), in cases where the data relevant to the analysis were not available in the published report, attempts were made to contact the authors of the individual studies, or data from other credible sources (e.g. trial registries) were used to extract the complete dataset.

Risk of bias and overall quality of evidence

Quality assessment for RCTs was evaluated according to the revised Cochrane Risk of Bias tool 2.0 (Sterne *et al.*, 2019) by two independent reviewers (K.I.K. and C.A.V.). The overall quality of evidence was graded according to the Grading of Recommendations Assessment, Development and Evaluation (GRADE) Working Group guidelines, using the GRADEpro tool (GRADEPro GDT, 2024).

Endpoints for meta-analysis

The main outcome measures were LBR per randomized patient (defined as the delivery of a single viable infant from 22 weeks of gestation onwards; deliveries of both singletons and multiples were registered as one delivery (Zegers-Hochschild *et al.*, 2017), using data only from the first cycle), ongoing pregnancy (at least one foetus with positive cardiac activity at 10–12 weeks of gestation), and clinical pregnancy (a viable pregnancy at 5–8 weeks of gestation). Cumulative live birth, ongoing pregnancy, and clinical pregnancy (defined as above) per oocyte collection were also evaluated by including all embryo transfers (either fresh or frozen) originating from a single attempted/planned oocyte retrieval until a live birth, ongoing pregnancy, or clinical pregnancy, respectively, was achieved or all embryos were used, whichever occurred first.

Secondary outcomes were miscarriage, multiple pregnancy, ectopic pregnancy, ovarian hyperstimulation syndrome (OHSS) (moderate to severe), total dose of gonadotrophins, duration of gonadotrophin administration, number of cumulus-oocyte complexes (COCs) retrieved, fertilization rates, number of embryos, and number of good and/or top-quality embryos. Miscarriage, ectopic pregnancy, and multiple pregnancy were defined as in The International Glossary on Infertility and Fertility Care, 2017 (Zegers-Hochschild *et al.*, 2017).

Statistical analysis

The dichotomous data results for each of the eligible studies for meta-analysis were expressed as risk ratio (RR) and the continuous data as mean difference with 95% CIs. Data were analysed according to the intention-to-treat principle. These results were combined for meta-analysis using the Mantel/Haenszel model when using the fixed effects model and the restricted maximum likelihood method with Hartung-Knapp-Sidik-Jonkman correction (Hartung and Knapp, 2001; IntHout *et al.*, 2014) when using the random effects model (in case of high heterogeneity, i.e. $I^2 \geq 50\%$). All results were combined for meta-analysis with the STATA Software (StataCorp. 2021, Stata Statistical Software: Release 17, College Station, TX, StataCorp LLC). Statistical heterogeneity was estimated with the I^2 statistic (Deeks *et al.*, 2019). Statistical significance was set at a P level of 0.05. Publication bias was explored using the Egger's test (Egger *et al.*, 1997).

A sensitivity analysis was performed for live birth, ongoing pregnancy, clinical pregnancy, and OHSS by excluding studies judged to be at overall high risk of bias according to RoB-2 (Sterne *et al.*, 2019).

Subgroup analyses on the primary outcome measure were planned according to different types of biosimilars, biosimilars approved by the FDA and/or EMA, and according to the type of patients (poor, normal, or high responders).

Results

Systematic review

Characteristics of the included studies

The literature search yielded 504 potentially relevant reports (Fig. 1). The titles of these manuscripts were examined, resulting in 124 potentially eligible publications. Subsequently, the abstracts of these studies were assessed, and 36 were identified as potentially suitable to provide data to answer the research question. The full text of these studies was then examined, resulting in the exclusion of 27 published reports (reasons for exclusion are presented in Fig. 1), and the final inclusion of 9 reports representing 8 RCTs (Rettenbacher *et al.*, 2015; Strowitzki *et al.*, 2016; Fertility Biotech AG, 2017; Barakhoeva *et al.*, 2019; Hu *et al.*, 2020; Pasqualini *et al.*, 2021; Rashidi *et al.*, 2021; Hu *et al.*, 2023a) published between 2015 and 2023. Seven biosimilar preparations were investigated, Bemfola/Afolia (two RCTs; NCT01121666, NCT01687712), Ovaleap (ISRCTN74772901), Cinnal-f (IRCT201011155181N1), Folitime (NCT02454556), QL1012 (NCT05149924), Primapur (NCT03088137), and Follitrope (NCT03506243). For the RCT NCT01687712, the data were obtained from ClinicalTrials.gov as no publication presenting the outcomes of this RCT was available in the databases screened. The Bemfola (NCT01121666) and Afolia (NCT01687712) trials had two and three stimulation cycles, respectively. However, only data from the first cycle were included, as per this systematic review and meta-analysis protocol. With regard to the Follitrope study (NCT03506243) (Hu *et al.*, 2020), a correction was published in 2023 (Hu *et al.*, 2023a) and was taken into consideration during the data extraction and synthesis.

Characteristics of the eight studies are presented in Table 1 and Supplementary Tables S2 and S3. Randomization method was reported in all the included studies, while allocation concealment was partially reported in the Afolia study (Fertility Biotech AG, 2017). Most of the studies (7/8) were single blinded for the outcome assessor, and only the Cinnal-f study (Rashidi *et al.*, 2021) was triple blinded (investigators, clinicians, patients). All studies were sponsored. The largest study was the Afolia by Fertility Biotech AG, published only on ClinicalTrials.gov (n = 1101). The risk of bias assessment of the eligible studies is presented in Supplementary Figs S1 and S2. Overall, three studies (Hu *et al.*, 2020; Pasqualini *et al.*, 2021; Hu *et al.*, 2023a) were deemed to be at high risk of bias, either due to deviations from the intended intervention (only per protocol data were presented

Table 1. Main methodological characteristics of the randomized controlled trials included in the meta-analysis.

Clinical trial registration number	Published report	Country	Recruitment period	Multi-centre (yes/no)	Sample size (reference product/biosimilars)	Power analysis (yes/no)	Method of randomization	Blinding	Allocation concealment	Primary outcome measure assessed	Financial support	Authors contacted (yes/no)
NCT01121666 Rettenbacher *et al.*, 2015 (Bemfola) Only fresh	Rettenbacher *et al.*, 2015, RBM Online	Multiple (EU)	July 2010–April 2012	Yes (15 centres, 6 countries)	372 (123/249)	Yes	Stratified by age (<35, ≥35) randomization using a minimization algorithm 2:1 ratio	Single (investigator/outcomes assessor)	Centralized allocation via an interactive web responses system	Number of oocytes retrieved	Sponsored by FINOX AG (currently Fertility Biotech)	No
ISRCTN74772901 Strowitzki *et al.*, 2016 (Ovaleap) Only fresh	Strowitzki *et al.*, 2016, RB&E	Multiple (EU)	March 2010–July 2011	Yes (20 centres, 5 countries)	299 (146/153)	Yes	Computer generated randomization stratified by using a block size of 2 1:1 ratio	Single (assessor, ie investigator and embryologist)	Via an interactive voice responses system	Number of oocytes retrieved	Sponsored by Teva Branded Pharmaceutical Products R&D, Inc.	No
NCT01687712 2017 (Afolia) Only fresh	Fertility Biotech Ag., ClinicalTrials.gov, accessed on 18 January 2024	USA	25 November 2013–10 September 2015	Yes (22 centres)	1101 (549/551) 1 not accounted for	Not reported	Centrally randomized in a 1:1 ratio	Single (investigator/outcomes assessor)	Central randomization	Clinical pregnancy 6 weeks post-embryo transfer	Sponsored by Fertility Biotech AG	No
NCT03088137 Barakhoeva *et al.*, 2019 (Primapur) Only fresh	Barakhoeva *et al.*, 2019, EJOGRB	Russia	8 February 2017–17 August 2018	Yes (3 centres)	110 (55/55)	Yes	Randomization based on the WinPepi random number generator programme, version 11.50 (module ETCETERA 3.26) at a ratio of 1:1	Single (outcomes assessor)	Centralized treatment allocation with an Interactive Web Response System (IWRS)	Number of oocytes retrieved	Sponsored by IVFarma LLC (Moscow, Russia)	Yes
NCT03506243 Hu *et al.*, 2020 (Follitrope®) Only fresh	Hu *et al.*, 2020 (+2023), Aging	China	April 2015 - May 2017	Yes (6 centres)	451 (112/339)	Yes	Randomization by random numbers generation using SAS 9.2 Software (SAS Institute, USA), stratified by age (20–30, 31–35, 36–39) in a 3:1 ratio	Single (outcomes assessor)	Sealed sequentially numbered envelopes	Number of cumulus-oophorous complexes	Sponsored by Hangzhou Yuyuan Bioscience Technology Co., Ltd	No
NCT05149924 Hu *et al.*, 2023a (QL1012) Only fresh	Hu *et al.*, 2023a, RBM Online	China	October 2018—June 2019	Yes (13 centres)	354 (176/178)	Yes	Central interactive web-response system stratified by age (<30, 30–34, and ≥35 years) with a block size of four, 1:1 ratio	Single (outcomes assessor)	Via an interactive web-response system	Number of oocytes retrieved	Sponsored by Qilu Pharmaceutical Co., Ltd	No
IRCT201011155181N1 Rashidi *et al.*, 2021 (Cinnal-f) Only fresh	Rashidi *et al.*, 2021, International Journal of Reproductive BioMedicine	Iran	April 2011–April 2013	No	200 (100/100)	Yes	Permuted block randomization, 1:1 ratio	Triple blinded (investigators, clinicians, patients)	3rd person handling the randomization list	Percentage of MII oocytes	Sponsored by CinnaGen Co. & Alborz University of Medical Sciences (Iran)	No
NCT02454556 Pasqualini *et al.*, 2021 (Folitime) Only fresh	Pasqualini *et al.*, 2021, JBRA Assisted Reproduction	Argentina	28 April 2015–6 September 2017	Yes (4 centres)	100 (50/50)	Yes	Computer algorithm, 1:1 ratio	Single (outcomes assessor)	Central allocation through the EDC (authors contacted)	Number of oocytes retrieved	Gema Biotech SAU (authors contacted)	Yes

Table 2. Population characteristics and details of ART treatment protocol used of the randomized controlled trials included in the meta-analysis.

Clinical trial registration number	Inclusion criteria	Exclusion criteria	Number of cycles	GnRH protocol used	Type of follitropin alfa	Day of initiation	Starting dose (IU)	Dose adaption criteria	Criteria for hCG administration	Trigger type and dose	Ovum pick-up	IVF/ ICSI	Day of ET	Luteal phase support protocol (type/duration)
NCT01121666 Rettenbacher *et al.*, 2015 **(Bemfola®)**	Age 20–38 years; BMI 18–30 kg/m^2; regular menstrual cycles of 25–35 days; first or second cycle in the present series of assisted reproduction technique; basal FSH <10 IU/L (cycle days 2–5); oestradiol levels <50 pg/ml on the first day of FSH administration; total antral follicle count of 10–25 follicles; infertility resulting from tubal factors; mild endometriosis ASRM stage 1–2; male factor; unexplained infertility; and presence of both ovaries and normal uterine cavity as confirmed by transvaginal ultrasound within 6 months before randomization	>2 previous ART retrieval cycles; the presence of endocrine disorder; known tumours of the hypothalamus and pituitary gland, or both; a history of severe ovarian hyperstimulation syndrome; severe endometriosis (ASRM stage 3 or 4); the presence of a hydrosalpinx; polycystic ovaries (Rotterdam criteria); history of poor response to gonadotrophin treatment (defined as <5 oocytes retrieved in a previous attempt); and any hormonal treatment within 1 month before starting FSH treatment (with the exception of levothyroxin)	2 cycles (no cross-over)	Long mid luteal GnRH Agonist (triptorelin or buserelin or leuprorelin)	Bemfola and GONAL-f	Successful downregulation (E2 level of <50 pg/ml, a shedded endometrium thickness <5 mm and no ovarian cysts)	150 IU	Adjusted after Day 6 if risk of OHSS or other safety concerns, decreased/ coasting/ terminated	≥1 follicle ≥18 mm and 2 follicles ≥16 mm	r-hCG (250 µg)	34–36 hours	IVF/ICSI (according to the centers standard procedures)	Day 2–5	Intravaginal utrogestan 3 × 200 mg/day from day of ET until confirmation of clinical pregnancy (5–6 weeks after oocyte retrieval)
ISRCT-N74772901 Strowitzki *et al.*, 2016 **(Ovaleap)**	Age 18–37 years; infertile (otherwise healthy); normogonadotrophic, 2 confirmed normal ovaries, undergoing controlled ovarian stimulation with ART therapy; regular menstrual cycles of 21–35 days; BMI 18–29 kg/m^2; basal FSH, estradiol, prolactin, and TSH concentrations in the normal range; able to give written informed consent	>2 previously completed consecutive unsuccessful IVF cycles or >3 miscarriages; history of severe OHSS; primary ovarian failure; categorized as poor responders to ovarian stimulation; hypersensitivity or allergy to recombinant FSH preparations; women with any significant cardiovascular, pulmonary, neurologic, endocrine, hepatic, or renal disease; neoplasm or a history of chemotherapy or radiation therapy; use of clomiphene or gonadotrophins within 30 days prior to enrolment.	1 cycle treatment	Long mid luteal GnRH Agonist (buserelin)	Ovaleap and GONAL-f	Successful downregulation (E2 < 50 pg/ml, negative pregnancy test and no ovarian cysts >10 mm)	150 IU	Adjusted after Day 5 to maximum 450 IU/day (based on follicle count size and appearance, serum oestradiol and ultrasound examination), every 3–5 days in increments of 37.5 IU, no more than 150 IU	≥3 follicles ≥17 mm and E2 levels below 5500 pg/mL	r-hCG (dose not reported)	34–37 hours	Not reported	Not reported	According to investigators discretion
NCT01687712 2017 **(Afolia)**	Age 35–42 years; Indication for IVF or ICSI; Regular menstrual cycles (25–35 days); History ≤2 fresh cycle treatments in the present series of ART at the day of first screening (thawed cycles are not subject to that criteria); BMI ≥18 ≤ 38 kg/m^2; Basal FSH <12 IU/L (cycle day 2–	Presence of pregnancy; PCOS; AFC >20 follicles with a diameter of <11 mm in menstrual cycle day 2–5; >2 previous unsuccessful fresh ART retrieval cycles; Uncontrolled endocrine disorder; Previous history or presence of severe OHSS; Intrauterine	Maximum of 3 cycles	Not reported	Afolia and GONAL-f RFF	On the day of successful downregulation	225 IU	Adjusted after Day 5 to maximum 450 IU/day or minimum 75 IU/day (in increments of 37.5 IU)	Not reported	hCG (dose and type not reported)	34–36 hours	IVF/ICSI	Not reported	Not reported

(continued)

Table 2. (Continued)

Clinical trial registration number	Inclusion criteria	Exclusion criteria	Number of cycles	GnRH protocol used	Type of follitropin alfa	Day of initiation	Starting dose (IU)	Dose adaption criteria	Criteria for hCG administration	Trigger type and dose	Ovum pick-up	IVF/ ICSI	Day of ET	Luteal phase support protocol (type/duration)
	5); AFC 10–20 follicles with a diameter of <11 mm in menstrual cycle day 2–5; infertility due to any of the following factors: tubal factor, mild endometriosis (ASRM stage 1–2), male factor, unexplained infertility; presence of both ovaries by ultrasonography and normal uterine cavity (confirmed by hysterosalpingography, saline infusion sonography or hysteroscopy within 6 months before randomization); male partner with semen analysis that is at least adequate for ICSI within 6 months prior to patient beginning down-regulation (invasive or surgical sperm retrieval, donor and/or cryopreserved sperm may be used); willingness to participate in the study and to comply with the study protocol; signed informed consent prior to screening	fibroids ≥5 cm or otherwise clinically relevant pathology that could impair embryo implantation or pregnancy continuation; History of recurrent spontaneous abortion (3 or more, even when unexplained); Presence of severe endometriosis (ASRM stage 3 or stage 4) or hydrosalpinx; Neoplasia, including tumors of the hypothalamus and pituitary gland; Abnormal bleeding of undetermined origin; History of extra-uterine pregnancy in the previous 3 months; Known allergy or hypersensitivity to progesterone or to any of the excipients (including peanut oil) of the additional study medications (GnRH agonist, Ovidrel®, and Crinone 8%®); History of poor response to gonadotrophin treatment (defined as fewer than 5 oocytes retrieved in a previous attempt); Any hormonal treatment within 1 month before the start of the FSH treatment, with the exception of levothyroxine); Egg donor; Administration of other investigational products within the previous month; Clinically abnormal findings at Visit 1; Concomitant participation in another study protocol												
NCT03088137 Barakhoeva *et al.*, 2019 **(Primapur)**	Age 20–35 years old with a regular menstrual cycle (duration: 21–35 days, established causes of infertility: tubal and/or male factors, first or second attempt at IVF/ICSI; ≥18 BMI ≤30 kg/m^2; FSH 10<IU/l and oestradiol level <50 pg/ml (cycle Days 2–5); AMH ≥ 1.0 ng/ml; 4 ≥AFC ≤15, written consent	Established contraindications to the use of ART methods; hypersensitivity to follitropin alfa or excipients; ≥2 cycles of IVF/ICSI; history of severe OHSS; PCOS; endometriosis; uterine cavity pathology; history of poor or excessive response to stimulation with a r-hFSH; male infertility factors: severe oligoasthenoteratozoospermia; azoospermia.	1 cycle	Flexible (antagonist initiated when leading follicle reached a mean diameter of 14 mm) GnRH Antagonist protocol (ganirelix)	Primapur and GONAL-f	Day 2–3 of menstrual cycle	150 IU	Adjusted after day 5 to maximum 450 IU/day based on ultrasound	Leading 2–3 follicles <18 mm (if at risk of OHSS >15 follicles <14 mm)	hCG (5000 - 10000 IU) or GnRH agonist (0.2 mg Decapeptyl)	<37 hours	IVF/ICSI	Day 3/5	According to investigators discretion

(continued)

Table 2. (Continued)

Clinical trial registration number	Inclusion criteria	Exclusion criteria	Number of cycles	GnRH protocol used	Type of follitropin alfa	Day of initiation	Starting dose (IU)	Dose adaption criteria	Criteria for hCG administration	Trigger type and dose	Ovum pick-up	IVF/ ICSI	Day of ET	Luteal phase support protocol (type/duration)
NCT03506243 Hu *et al.*, 2020 **(Follitrope®)**	Age 20–39 years; regular menstrual cycle (duration: 25–35 days; normal baseline FSH, LH, oestradiol and progesterone levels; indication for IVF: tubal factor, unexplained infertility, male infertility, combined factors; written consent	Presence of clinically significant major systemic disease or endocrine or metabolic disorders; BMI >30 kg/m^2; Presence of abnormalities of the uterus (uterine fibroids [≥2 cm in diameter or affecting the endometrium], endometrial polyp, intrauterine adhesion, uterine malformation), ovary (polycystic ovary, ovarian cyst), or appendix (hydrosalpinx) at the time of randomization; A medical history of uterine/ovarian surgery; A history of ovarian hyperstimulation syndrome (OHSS) in previous IVF cycles; Poor response to previous gonadotropin therapy; * Presence of the following: previous poor ovarian response (<3 oocytes through previous conventional stimulation protocol), Abnormal ovarian reserve testing (ie, <5 antral follicles); Abnormal uterine bleeding of unknown cause; At least one member of the couple will receive donor sperms or donor eggs, or preimplantation genetic diagnosis; positive for HIV or syphilis; presence of known serious mental disorders or inability to understand the purpose, methods, and other aspects of this clinical study and to follow the study procedures; A habit of alcoholism, heavy smoking, and/or drug abuse; A history of allergy to recombinant human follicle-stimulating hormone (r-hFSH) or any of its excipients; Subjects who are found to participate in any other	1 cycle	Long mid luteal GnRH Agonist (triptorelin)	Follitrope and GONAL-f	Successful downregulation (median antral follicle size <10 mm)	150 IU/day aged 20–30 150–225 IU/day aged 31–35 225–300 IU/day aged >35	Adjusted from day 5 based on ovarian response (incremental adjustment in dose of 75 IU was permitted with a maximum daily dose of 450 at an interval of 3 days)	≥1 follicles ≥18 mm or ≥3 follicles ≥16 mm	hCG (5000–10000 IU)	36–38 hours	IVF/ICSI	Day 3	IM progesterone 60 mg/day for 10 weeks after ET

(continued)

Table 2. (Continued)

Clinical trial registration number	Inclusion criteria	Exclusion criteria	Number of cycles	GnRH protocol used	Type of follitropin alfa	Day of initiation	Starting dose (IU)	Dose adaption criteria	Criteria for hCG administration	Trigger type and dose	Ovum pick-up	IVF/ ICSI	Day of ET	Luteal phase support protocol (type/duration)
		clinical study following enrollment in this study or who have participated in another clinical study 3 months prior to randomization in this study; A medical history of ovarian, breast, uterine, or hypothalamic or pituitary tumors; More than 3 previous consecutive failed IVF cycles; Clomiphene or gonadotropin therapy within 1 month prior to randomization; Patients with positive results in serum pregnancy test; Failure to reach the standard (antral follicles size ≥10 mm) after 14 ± 2 days of downregulation												
NCT05149924 Hu *et al.*, 2023a **(QL1012)**	Age 20–39 years; body mass index 18– 30 kg/m^2; regular menstrual cycles 25–35 days; and basal serum FSH, luteinizing hormone (LH), oestradiol and progesterone in the normal range or clinically irrelevant abnormal range; and had indication for IVF/ ICSI-ET	History of three or more unsuccessful IVF/ICSI-ET cycles; three or more previous spontaneous miscarriages; the presence of a high risk of ovarian hyperstimulation syndrome (OHSS); poor ovarian function; a history of malignancy; and treatment with any fertility regulating or hormonal agents within 1 month before randomization	1 cycle	Long mid luteal GnRH Agonist (triptorelin)	QL1012 and GONAL-f	Successful downregulation (LH <5 IU/l and oestradiol <50 pg/ml)	75–150 IU <30 years 150–225 IU for ages 30–34 225–300 IU ≥ 35 years	Adjusted from day 5 to maximum 450 IU/ day (in increments of 37.5 IU no more than 150 IU) according to follicular development and serum oestradiol	≥1 follicle >18 mm or ≥2 follicles >16 mm	250μg HCG alpha (Ovitrelle) with or without HCG 2000 IU	36–38 hours	IVF/ICSI	Day 3/5	Vaginal progesterone gel (Crenome) 90 mg/daily and dydrogesterone p. os 10 mg (Duphaston) twice daily
IRC-T201011155-181N1 Rashidi *et al.*, 2021 **Cinnal-f**	Age < 35 years; FSH levels ≤8 mIU/ml on the second day of the cycle; regular menstrual cycles of 25–35 days; and receiving gonadotropin-releasing hormone (GnRH) agonist for pituitary suppression	History of previous ovarian surgery or hyperstimulation syndrome; had any endocrinopathy or an abnormal hormonal profile; smoked cigarettes or were allergic to Gonal-f®, Cinnal-f® or any of their components	1 cycle	Long mid luteal GnRH Agonist Agonist (buserelin)	Cinnal-f and GONAL-f	Day 2 of menstrual cycle	150 IU	Day 7 or 8 based on ultrasound scan	> follicle ≥18 mm and ≥3 follicles ≥14 mm	hCG 10.000 IU	36 hours	ICSI	Day 2/3	Vaginal progesterone 400 mg/ daily (Cyclogest)
NCT02454556 Pasqualini *et al.*, 2021 **(Folitime)**	Age 18–38 years, BMI 18–32 kg/m2, regular menstrual cycles of 25–35 days, basal FSH < 10 IU/L (cycle day 2–5), undergoing their first or second	More than two previous unsuccessful retrieval cycles, history of ovarian hyperstimulation syndrome (OHSS); moderate to severe endometriosis; presence of hydrosalpinx;	1 cycle	Fixed (day 6) GnRH Antagonist (cetrorelix)	Folitime and GONAL-f	Day 2–3 of menstrual cycle	225 IU	Adjusted after day 5 to maximum 450 IU/ day based on ultrasound (every 3 days in	≥3 follicles ≥17 mm	uHCG 10.000 IU	Not reported	IVF/ICSI	Day 3/5	Vaginal progesterone (utrogestan) 600 mg daily

(continued)

Table 2. (Continued)

Clinical trial registration number	Inclusion criteria	Exclusion criteria	Number of cycles	GnRH protocol used	Type of follitropin alfa	Day of initiation	Starting dose (IU)	Dose adaption criteria	Criteria for hCG administration	Trigger type and dose	Ovum pick-up	IVF/ ICSI	Day of ET	Luteal phase support protocol (type/duration)
	cycle in the present series of IVF, total AFC ≥ 8 to ≤18 follicles, with a follicle diameter <10 mm (sum of both ovaries) in the early follicular phase; documented history of infertility due to tubal factor, male factor, or unexplained infertility; presence of both ovaries and a normal uterine cavity as confirmed by transvaginal ultrasound within three months of randomization	history of poor ovarian response to gonadotropin treatment (defined as fewer than five oocytes retrieved in a previous attempt); any hormonal treatment within one month of the start of the FSH treatment (with the exception of levothyroxine); intrauterine leiomyomas ≥5 cm; ovarian cyst or enlargement of undetermined origin; otherwise clinically relevant disease; and history of recurrent miscarriage (three or more, even when unexplained) or extrauterine pregnancy in the previous three months						adjustments of 75–150 IU)						

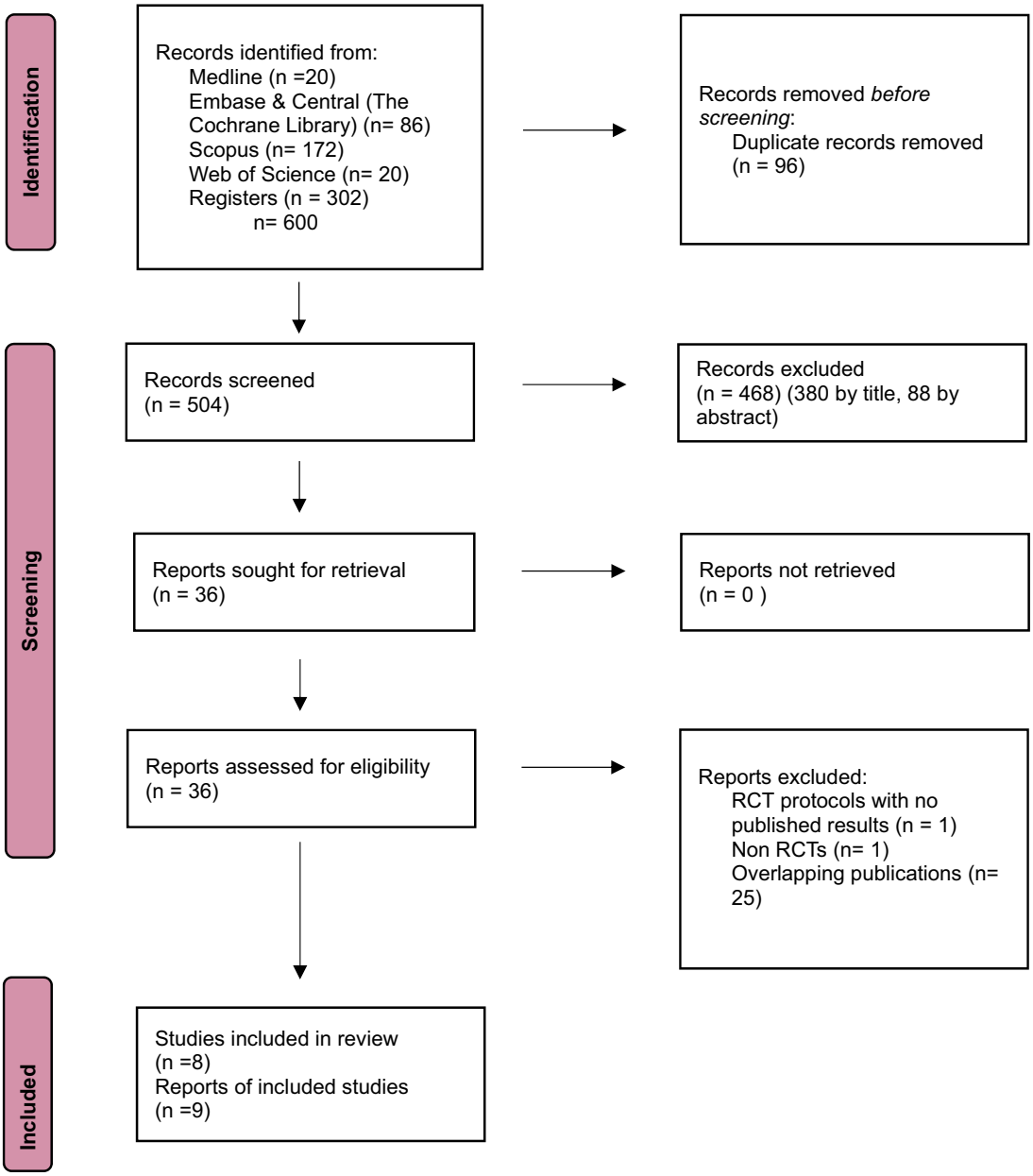


Figure 1. PRISMA flow chart. For more information, visit: http://www.prisma-statement.org/. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-analysis; RCT, randomized clinical trials.

regarding live birth, the primary outcome of this meta-analysis) (Pasqualini *et al.*, 2021; Hu *et al.*, 2023a) or due to deviations from the intended intervention and missing outcome data (Hu *et al.*, 2020) (only per protocol data were presented and >30% of randomized patients did not have an embryo transfer).

Four out of eight studies excluded patients who previously had severe OHSS (Rettenbacher *et al.*, 2015; Strowitzki *et al.*, 2016; Fertility Biotech AG, 2017; Barakhoeva *et al.*, 2019) and the remaining studies excluded patients who had a previous history of any type of OHSS (Hu *et al.*, 2020; Pasqualini *et al.*, 2021; Rashidi *et al.*, 2021; Hu *et al.*, 2023a) (Table 2). All RCTs excluded women with a history of poor ovarian response except from the Cinnal-f trial (Rashidi *et al.*, 2021). Based on the information provided, five out of eight studies (Rettenbacher *et al.*, 2015; Fertility Biotech AG, 2017; Barakhoeva *et al.*, 2019; Hu *et al.*, 2020; Pasqualini *et al.*, 2021) were deemed to have been performed in normal responders, a single study (Strowitzki *et al.*, 2016) was performed in infertile patients excluding poor responders and the provided data were not sufficient to confidently ascertain the type of population for the remaining two studies (Rashidi *et al.*, 2021; Hu *et al.*, 2023a). None of the studies was explicitly performed on poor or high responders.

Additional information were obtained after contacting the authors of the Folitime (Pasqualini *et al.*, 2021) and the Primapur (Barakhoeva *et al.*, 2019) study. Any other attempts to contact the authors from the remaining studies were unsuccessful (either no reply was received or corresponding authors' contact details were no longer valid).

Meta-analysis

Primary outcome measure

Live birth

A significantly lower LBR was observed using the biosimilars of follitropin alfa (Rettenbacher *et al.*, 2015; Strowitzki *et al.*, 2016; Fertility Biotech AG, 2017; Barakhoeva *et al.*, 2019; Pasqualini

et al., 2021; Hu et al., 2023a) compared to the originator in women undergoing ovarian stimulation for ART (RR: 0.83, 95% CI: 0.72–0.96; I^2 = 0%, six studies, n = 2335; moderate certainty of evidence) (Fig. 2A). Publication bias did not seem to be present ($P = 0.21$).

The sensitivity analysis, which excluded two studies (Pasqualini et al., 2021; Hu et al., 2023a) due to high risk of bias, did not alter the effect size (RR: 0.83, 95% CI: 0.71–0.97; I^2 = 0%, four studies, n = 1881; moderate certainty of evidence) (Supplementary Fig. S3). The subgroup analyses performed did not detect a statistically significant difference between subgroups when studies were grouped by types of biosimilars (Afolia/Bemfola/Ovaleap, Primapur, QL1012, Cinnal-f, Folitime, Follitrope) ($P = 0.518$), by whether the biosimilar had received FDA and/or EMA approval (Afolia/Bemfola/Ovaleap) or not (Primapur, QL1012, Cinnal-f, Folitime, Follitrope) ($P = 0.668$), and the type of population (normal responders: Bemfola, Afolia, Primapur, Follitrope and Folitime versus unclassified: Ovaleap, Cinnal-f, QL1012) ($P = 0.648$).

Secondary outcomes

Ongoing pregnancy

A significantly lower ongoing pregnancy rate was observed using the biosimilars of follitropin alfa (Rettenbacher et al., 2015; Strowitzki et al., 2016; Barakhoeva et al., 2019; Hu et al., 2020; Pasqualini et al., 2021; Rashidi et al., 2021; Hu et al., 2023a) compared to the originator in women undergoing ovarian stimulation for ART (RR: 0.81, 95% CI: 0.70–0.94; I^2 = 0%, seven studies, n = 1886; low certainty of evidence) (Fig. 2B). Publication bias did not seem to be present ($P = 0.43$).

The sensitivity analysis, which excluded three studies (Hu et al., 2020; Pasqualini et al., 2021; Hu et al., 2023a) due to high risk of bias, did not alter the effect size (RR: 0.80, 95% CI: 0.67–0.97; I^2 = 0%, four studies, n = 981; moderate certainty of evidence) (Supplementary Fig. S4).

Clinical pregnancy

A significantly lower clinical pregnancy rate was observed using the biosimilars of follitropin alfa (Rettenbacher et al., 2015; Strowitzki et al., 2016; Fertility Biotech AG, 2017; Hu et al., 2020; Pasqualini et al., 2021; Rashidi et al., 2021; Hu et al., 2023a) compared to the originator in women undergoing ovarian stimulation for ART (RR: 0.82, 95% CI: 0.73–0.92; I^2 = 0%, seven studies, n = 2876; low certainty of evidence) (Fig. 2C). Publication bias did not seem to be present ($P = 0.46$).

The sensitivity analysis, which excluded three studies (Hu et al., 2020; Pasqualini et al., 2021; Hu et al., 2023a) due to high risk of bias, did not alter the effect size (RR: 0.81, 95% CI: 0.70–0.93; I^2 = 0%, four studies, n = 1971; moderate certainty of evidence) (Supplementary Fig. S5).

Ovarian hyperstimulation syndrome

In view of very limited data for moderate and severe OHSS (Strowitzki et al., 2016; Fertility Biotech AG, 2017; Pasqualini et al., 2021), OHSS rate was calculated encompassing all the cases reported, regardless of their severity. No difference was observed in OHSS rates (RR: 1.17, 95% CI: 0.90–1.52; I^2 = 0%, eight studies, n = 2986) (Rettenbacher et al., 2015; Strowitzki et al., 2016; Fertility Biotech AG, 2017; Barakhoeva et al., 2019; Hu et al., 2020; Pasqualini et al., 2021; Rashidi et al., 2021; Hu et al., 2023a) (Fig. 3A) between the two groups. Publication bias did not seem to be present ($P = 0.74$).

The sensitivity analysis, which excluded three studies (Hu et al., 2020; Pasqualini et al., 2021; Hu et al., 2023a) due to high risk of bias, did not alter the effect size (RR: 1.14, 95% CI: 0.72–1.83; I^2 = 0%, five studies, n = 2081) (Supplementary Fig. S6).

Duration of gonadotrophin administration

A shorter duration of gonadotrophin administration was observed using the biosimilars of follitropin alfa (Ovaleap, Bemfola, Cinnal-f, Primapur, Afolia) (Rettenbacher et al., 2015; Strowitzki et al., 2016; Fertility Biotech AG, 2017; Barakhoeva et al., 2019; Rashidi et al., 2021) compared to the originator in women undergoing ovarian stimulation for ART (weighted-mean-difference [WMD]: –0.19 days, 95% CI: –0.34 to –0.05; I^2 = 0%, five studies, n = 2081) (Fig. 3B). Publication bias did not seem to be present ($P = 0.99$).

Total dose of FSH

No significant difference was observed in the total dose of FSH when using the biosimilars of follitropin alfa (Rettenbacher et al., 2015; Strowitzki et al., 2016; Fertility Biotech AG, 2017; Barakhoeva et al., 2019; Rashidi et al., 2021) as compared to the originator in women undergoing ovarian stimulation for ART (WMD: –34.69 IUs, 95% CI: –74.54 to 5.16; I^2 = 15.53%, five studies, n = 2081) (Fig. 3C). Publication bias did not seem to be present ($P = 0.16$).

Number of COCs retrieved

No difference was observed in the number of COCs retrieved using the biosimilars of follitropin alfa (Rettenbacher et al., 2015; Strowitzki et al., 2016; Barakhoeva et al., 2019; Hu et al., 2020; Pasqualini et al., 2021; Rashidi et al., 2021) compared to the originator in women undergoing ovarian stimulation for ART (WMD: 0.27, 95% CI: –0.43 to 0.96; I^2 = 10.71%, six studies, n = 1527) (Fig. 3D). Publication bias did not seem to be present ($P = 0.42$).

Fertilization rates

Only one study provided data on fertilization rates per intention to treat (Rettenbacher et al., 2015) and therefore no meta-analysis was performed. Based on the results of that single study, no significant differences in fertilization rates between the two groups were detected (Biosimilar mean: 66.1%, SD: 24.84 versus Originator mean: 64%, SD: 24.76).

Multiple pregnancy

No difference in multiple pregnancy rates were observed, based on pooling of data from three studies (Strowitzki et al., 2016; Barakhoeva et al., 2019; Pasqualini et al., 2021) (RR: 1.31, 95% CI: 0.62–2.78; I^2 = 0%, three studies, n = 509). Publication bias did not seem to be present ($P = 0.76$).

Ectopic pregnancy

No differences in ectopic pregnancy rates were observed based on pooling of data from three studies (Strowitzki et al., 2016; Fertility Biotech AG, 2017; Barakhoeva et al., 2019) (RR: 1.15, 95% CI: 0.40–3.26; I^2 = 0%, three studies, n = 1509). Publication bias did not seem to be present ($P = 0.82$).

Miscarriage

No differences in miscarriage rates were observed based on pooling of data from two studies (Rettenbacher et al., 2015; Strowitzki et al., 2016) (RR: 0.91, 95% CI: 0.45–1.82; I^2 = 0%, two studies, n = 671). Publication bias did not seem to be present ($P = 0.83$).

Other outcomes

Data per randomized patient on the mean number of embryos and mean number of good and/or top-quality embryos were not

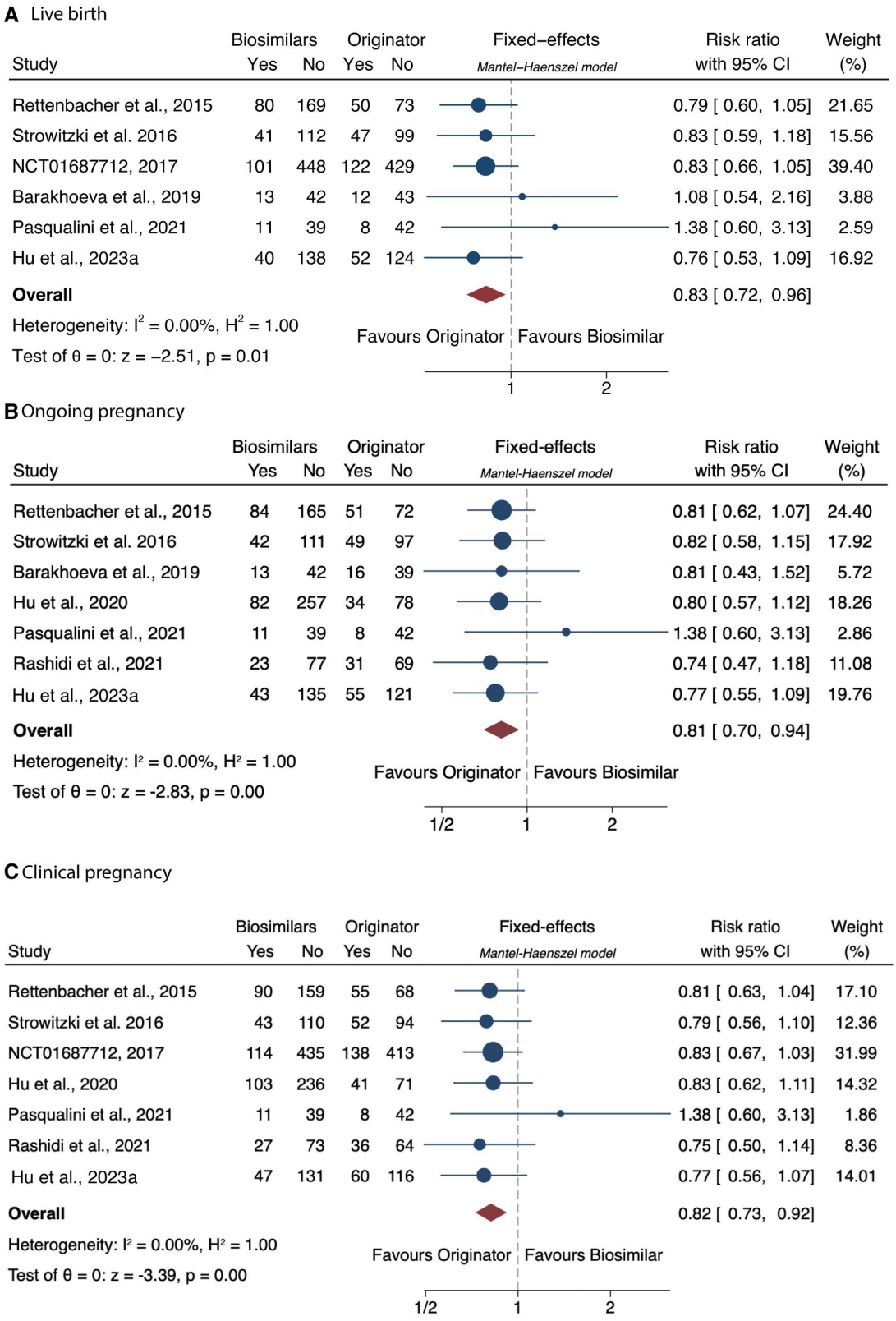
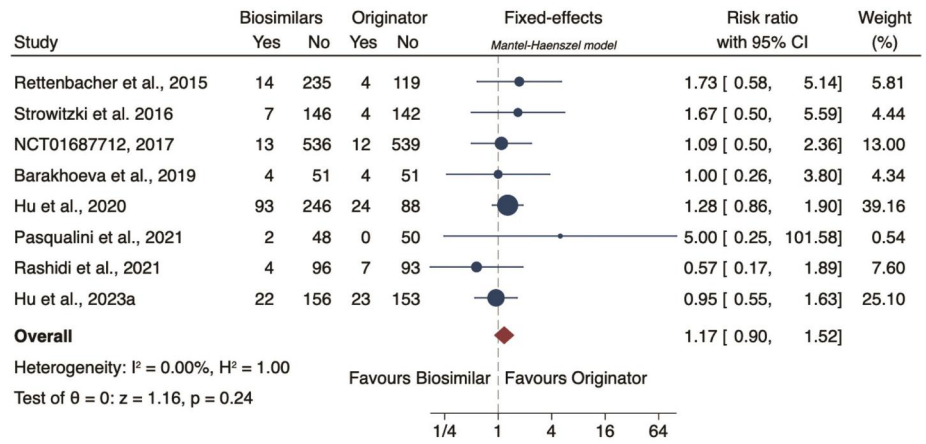
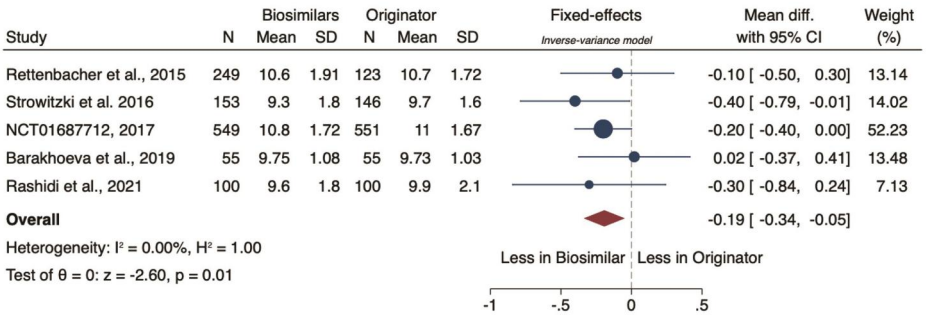


Figure 2. Forest plots, presenting the results of the meta-analysis on pregnancy outcomes. (**A**): Forest plot comparing live birth rates between patients stimulated with biosimilars of follitropin alfa and those who were stimulated with the originator; (**B**): Forest plot comparing ongoing pregnancy rates between patients stimulated with biosimilars of follitropin alfa and those who were stimulated with the originator; (**C**): Forest plot comparing clinical pregnancy rates between patients stimulated with biosimilars of follitropin alfa and those who were stimulated with the originator.

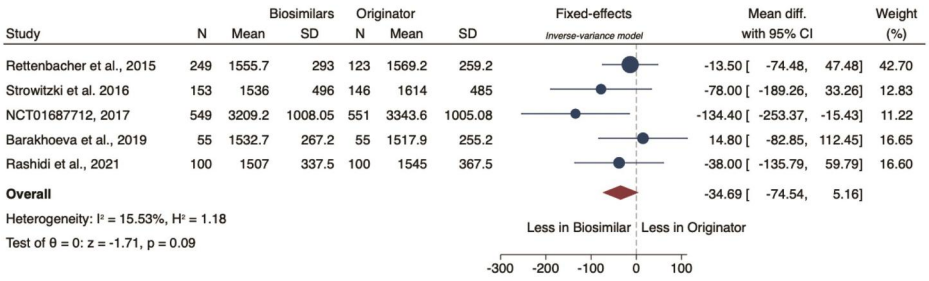
A Ovarian hyperstimulation syndrome (OHSS)



B Duration of gonadotrophin administration (days)



C Total dose of FSH (IUs)



D Number of oocytes retrieved

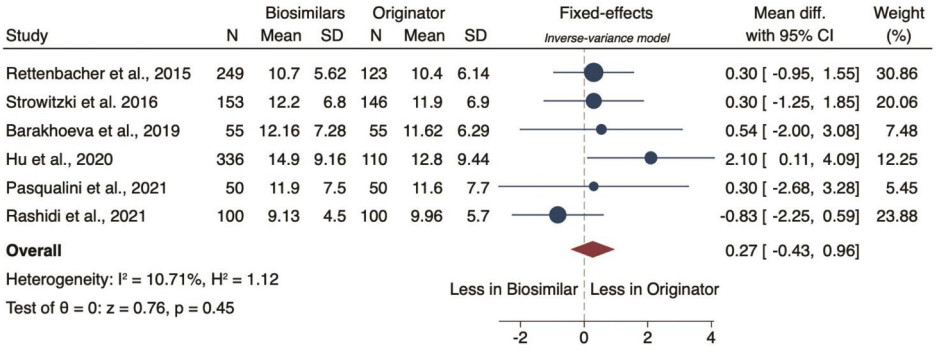


Figure 3. Forest plots, presenting the results of the meta-analysis on secondary, non-pregnancy outcomes. (**A**): Forest plot comparing ovarian hyperstimulation syndrome rates between patients stimulated with biosimilars of follitropin alfa and those who were stimulated with the originator; (**B**): Forest plot comparing the duration of gonadotrophin administration (in days) between patients stimulated with biosimilars of follitropin alfa and those who were stimulated with the originator; (**C**): Forest plot comparing the total dose of FSH between patients stimulated with biosimilars of follitropin alfa and those who were stimulated with the originator; (**D**): Forest plot comparing the number of oocytes retrieved between patients stimulated with biosimilars of follitropin alfa and those who were stimulated with the originator.

provided in the eligible studies, and, therefore, a meta-analysis on these outcomes was not feasible.

A meta-analysis on cumulative live birth, ongoing pregnancy, and clinical pregnancy rates per oocyte collection (encompassing all embryo transfers originating from a single attempted/planned oocyte retrieval, required to reach a live birth, ongoing pregnancy, or clinical pregnancy, respectively) was also not feasible due to the lack of relevant data from the eligible studies.

Discussion

This systematic review and meta-analysis of eight RCTs including ~3000 patients suggest a relative reduction of 17% in the probability of live birth when using biosimilars compared to using the originator of follitropin alfa (n = 6 studies, 2335 patients). This finding was persistent in the sensitivity analysis performed, which excluded studies at high risk of bias. Moreover, the meta-analysis demonstrated a similarly lower probability with biosimilar preparations of follitropin alfa compared to the originator in ancestor outcomes, such as ongoing and clinical pregnancy, suggesting that there is no differential effect beyond the clinical pregnancy stage.

The most recent systematic review and meta-analysis published on this subject (Chua et al., 2021) has reviewed and pooled data from five RCTs published up to October 2020, all of which have been Phase III clinical trials evaluating biosimilars of follitropin alfa. Regarding live birth, their analysis eventually pooled data from four RCTs and detected a significantly decreased rate in the biosimilars group when compared to the originator (RR: 0.83, 95% CI: 0.71–0.97; four RCTs, n = 1881, I^2 = 0%). Moreover, the study suggested, ongoing pregnancy, and clinical pregnancy rates were significantly lower with biosimilars of follitropin alfa compared with the originator. At the same time, that meta-analysis associated the biosimilars with a shorter duration of gonadotrophin administration, leading to a higher number of COCs retrieved.

The outcomes of the current systematic review are consistent with those reported in the aforementioned publication, and the same effect estimate has been maintained with increased precision (narrower CI). Two more RCTs (Pasqualini et al., 2021; Hu et al., 2023a) have been included in the analysis for the primary outcome (RR: 0.83, 95% CI: 0.72–0.96; I^2 = 0%, 6 studies, n = 2335) increasing the total sample size by ~20%. Similarly, the statistical syntheses of seven studies for clinical and ongoing pregnancy were both in favour of the originator and were not materially altered in the sensitivity analyses performed. In contrast with the previously published systematic review and meta-analysis, the current one did not detect a difference in the number of oocytes retrieved between the biosimilars and the originator of follitropin alfa. Nevertheless, a recent corrigendum (Venetis et al., 2023) supported that the difference in the number of oocytes retrieved between the originator and the biosimilars of follitropin alfa was the result of incorrectly presented data in the original manuscript (Hu et al., 2020), and this was confirmed by an issued correction by the authors of the original article (Hu et al., 2023b). Incorporating the correct figures in the original meta-analysis showed that a significant difference in the number of oocytes retrieved was never actually present (Venetis et al., 2023).

An important consideration when assessing the results of this meta-analysis is the considerable homogeneity between the eligible studies. Even though these studies evaluated seven different biosimilars, the population and the clinical protocols used in these trials were similar. This was also confirmed by the fact that the statistical heterogeneity, as measured by the I^2, was negligible. Importantly though, the effect sizes on live birth, ongoing pregnancy, and clinical pregnancy observed in the six largest individual RCTs are very similar to the pooled effect size. The two RCTs (Barakhoeva et al., 2019; Pasqualini et al., 2021) where the effect size is not suggestive of lower pregnancy rates in the biosimilars group are the two smallest ones with significant uncertainty as expressed by wide 95% CIs.

Furthermore, a recently published study using real-world evidence (RWE) from the French National Health System comparing the outcomes of different types of gonadotrophins also suggested that biosimilars of follitropin alfa were associated with 19% lower odds of live birth (adjusted odds ratio: 0.81, 95% CI: 0.76–0.86) compared to the originator (Grynberg et al., 2023). This large (>240,000 cycles analysed) study, which statistically controlled for a number of potential confounders, including female age, number of previous oocyte pick-ups, and parity, also suggested lower odds of cumulative live birth with biosimilars of follitropin alfa compared to the originator (adjusted hazard ratio: 0.86, 95% CI: 0.82–0.89). This agreement between two different sources of evidence (meta-analysis of RCTs and RWE) is important, as it essentially triangulates the evidence and adds confidence to the findings (Heale and Forbes, 2013).

The explanation of the differences in pregnancy rates despite producing a similar yield of oocytes remains elusive. The most likely explanations are a differential effect on oocyte quality or endometrial receptivity. A discrepancy in the quality of the oocyte could be the result of variations between these products in terms of glycosylation and the resulting isoforms (Wang et al., 2016; Lunenfeld et al., 2019; Dias and Ulloa-Aguirre, 2021) or even in terms of batch-to-batch variability. Such differences between the originator of follitropin alfa and some of its biosimilars (i.e. Bemfola and Ovaleap) have already been demonstrated and could provide a pharmacological basis for the observed clinical differences (Mastrangeli et al., 2017; Riccetti et al., 2019; Manzi et al., 2022). It should be noted that biosimilars are associated with a shorter duration of stimulation, without any detectable differences in the total dose of FSH and the number of oocytes retrieved. This difference in the stimulation pattern might be associated with differences in oocyte quality and therefore with the observed differences in LBRs. These hypotheses, however, could not be explored as data on embryo quality were not available for most studies and/or could not be statistically synthesized.

An alternative hypothesis could be a negative effect of biosimilars on the endometrium either directly or via increased steroids (e.g. late follicular progesterone), which could impact negatively on endometrial receptivity (Bosch et al., 2010; Venetis et al., 2013; Venetis et al., 2015; Venetis et al., 2016). Data on late follicular steroids, and particularly serum progesterone, were not unfortunately available and therefore this hypothesis could also not be explored.

Finally, it has been argued by some (de Mora and Howles, 2023) that the observed differences in pregnancy rates might be attributable to product-unrelated factors such as the hCG dose given to trigger final oocyte maturation, the time interval between oocyte collection and the laboratory procedures followed or even the individual embryo transfer operator. However, such a hypothesis is highly problematic as these are rigorously designed RCTs with strict protocols intended to be used for regulatory approval. Importantly, all included RCTs have been designed, executed, and/or sponsored by entities that were interested in obtaining regulatory approval and commercially benefiting from the biosimilars. Therefore, the scenario of consistent

performance bias in favour of the originator product appears to be highly unlikely (Venetis and Mol, 2023).

On the other hand, there are some limitations that need to be considered. First, the meta-analysis includes RCTs from seven different biosimilars, which might not have the same pharmacodynamic profile. Nevertheless, the clinical data from the individual studies appear to be rather homogeneous and this was also tested through formal subgroup analyses, which failed to detect any significant differences. Another limitation of this meta-analysis is that only data regarding the fresh embryo transfer cycle were available, and, therefore, it is currently unknown whether there are differences in terms of cumulative live birth, ongoing pregnancy, and clinical pregnancy rates per aspiration. Robust data on oocyte or even better embryo quality would have been helpful in the interpretation of the findings of this meta-analysis, but unfortunately only three studies (Rettenbacher *et al.*, 2015; Pasqualini *et al.*, 2021; Hu *et al.*, 2023a) compared embryo quality (using different outcome measures) and none supported a difference between the biosimilars and the originator of follitropin alfa. Considering the heterogeneous way these comparisons were performed and that in all cases the analysis was not been performed 'per intention to treat' but 'per protocol', no reliable conclusions could be drawn (Supplementary Table S3). Finally, the population examined in the eligible RCTs does not explicitly include poor or high responders, and therefore, caution should be exercised before extrapolating these results to these populations.

Patients and physicians should be informed that biosimilars of follitropin alfa, although more affordable in many countries, are likely to be associated with reduced pregnancy rates compared to the originator. This could affect the cost-effectiveness of these products, which could lead to the originator being more cost-effective despite its higher overall cost (Schwarze *et al.*, 2022).

Moreover, as biosimilars are key products in the pharmaceutical industry, it is imperative that more research is targeted in confirming or rebutting the findings of this meta-analysis, and importantly elucidating the physiological mechanisms by which biosimilars might be affecting pregnancy outcomes. This would ideally result in the optimization and the production of biosimilars so that they lead to similar LBRs compared to the originator of follitropin alfa.

In conclusion, based on the best available evidence originating from eight RCTs, biosimilars of follitropin alfa are associated with similar number of oocytes but lower live birth, ongoing pregnancy, and clinical pregnancy rates compared to the originator.

Supplementary data

Supplementary data are available at *Human Reproduction* online.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Authors' roles

K.I.K.: Contributed to the construction of the protocol, performed the analyses and interpretation of the data, and drafted the manuscript. A.S.: Contributed to data extraction and the interpretation of the data and revised the manuscript for important intellectual content. E.M.K.: Reviewed the protocol of the study, contributed in the interpretation of the data, and revised the manuscript for important intellectual content. B.W.M.: Reviewed the protocol of the study, contributed in the interpretation of the data, and revised the manuscript for important intellectual content. C.A.V.: Conceived the idea of the study, contributed in the construction of the protocol, performed the analyses and interpretation of the data, and revised the manuscript for important intellectual content. All authors approved the final version of the manuscript.

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Conflict of interest

K.I.K. and A.S. have no competing interest to disclose. E.M.K. reports personal fees and non-financial support from Merck, Ferring, IBSA, and Vianex. B.W.M. has been supported by an investigator grant from NHMRC, has received consulting fees from Organon, Merck, and Norgine, research support and non-financial support from Merck KGaA, Darmstadt, Germany. B.W.M. also reports having stocks from OBsEva. C.A.V. reports grants, personal fees, and non-financial support from Merck KGaA, Darmstadt, Germany, personal fees and non-financial support from Merck, Sharpe and Dohme, personal fees and non-financial support from Organon, grants and non-financial support from Ferring, personal fees from IBSA, personal fees and non-financial support from Gedeon-Richter and Vianex.

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